

# **Cost-effectiveness of test-and-treat strategies to reduce the antibiotic prescription rate for acute febrile illness in primary healthcare clinics in Africa**

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**Potential conflicts of interest**

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**Table A1** Population characteristics of Accelerator trial per country

|                                  | <b>Burkina Faso</b>         |                                  | <b>Ghana</b>                |                                  | <b>Uganda</b>                |                                   |
|----------------------------------|-----------------------------|----------------------------------|-----------------------------|----------------------------------|------------------------------|-----------------------------------|
|                                  | Control<br>(%)<br>(n = 862) | Intervention<br>(%)<br>(n = 856) | Control<br>(%)<br>(n = 751) | Intervention<br>(%)<br>(n = 761) | Control<br>(%)<br>(n = 1211) | Intervention<br>(%)<br>(n = 1189) |
| Age<br>population<br>(years)     |                             |                                  |                             |                                  |                              |                                   |
| < 5                              | 649 (75)                    | 653 (76)                         | 570 (76)                    | 584 (77)                         | 377 (31)                     | 364 (31)                          |
| 5 – 9                            | 159 (18)                    | 146 (17)                         | 125 (17)                    | 118 (15)                         | 236 (19)                     | 196 (16)                          |
| 10 – 17                          | 54 (6)                      | 57 (7)                           | 56 (7)                      | 59 (8)                           | 169 (14)                     | 143 (12)                          |
| ≥ 18                             | -                           | -                                | -                           | -                                | 429 (35)                     | 486 (41)                          |
| Sex                              |                             |                                  |                             |                                  |                              |                                   |
| Female                           | 425 (49)                    | 415 (48)                         | 341 (45)                    | 348 (47)                         | 742 (61)                     | 717 (60)                          |
| Male                             | 437 (51)                    | 441 (52)                         | 410 (55)                    | 362 (53)                         | 469 (39)                     | 472 (40)                          |
| Presenting signs and<br>symptoms |                             |                                  |                             |                                  |                              |                                   |
| Respiratory                      | 398 (46)                    | 506 (59)                         | 412 (55)                    | 427 (58)                         | 421 (35)                     | 648 (54)                          |
| Non-<br>respiratory              | 464 (54)                    | 350 (41)                         | 339 (45)                    | 311 (42)                         | 790 (65)                     | 541 (46)                          |

**Table A2** Probabilities overview

&lt;Excel file&gt;

**Table A3** Overview of drug cost per country

| Drug name                 | Cost     |          | Source           |
|---------------------------|----------|----------|------------------|
|                           | Tube     | Per pill | Per prescription |
| <b>Burkina Faso</b>       |          |          |                  |
| albendazole               |          | \$ 0.24  | Local PI         |
| artesunate                | \$ 1.45  |          | \$ 1.45          |
| aspirin injectable        | \$ 1.45  |          | \$ 1.45          |
| carbocisteine             | \$ 4.84  |          | \$ 4.84          |
| chlorpheniramine          | \$ 0.48  |          | \$ 0.48          |
| coartem                   | \$ 2.72  |          | \$ 2.72          |
| diclofenac                | \$ 0.73  |          | \$ 0.73          |
| eosine                    | \$ 4.84  |          | \$ 4.84          |
| fluconazole               | \$ 6.29  | \$ 4.36  |                  |
| griseofulvin              | \$ 1.21  |          | \$ 1.21          |
| ibuprofen                 | \$ 2.90  | \$ 0.10  |                  |
| litacold syrup            | \$ 1.45  |          | \$ 1.45          |
| loperamide                |          | \$ 0.97  |                  |
| mebendazole               | \$ 1.69  | \$ 0.08  |                  |
| miconazole                | \$ 2.42  |          | \$ 2.42          |
| oral rehydration solution |          | \$ 0.51  |                  |
| paracetamol               | \$ 2.06  | \$ 0.10  |                  |
| metronidazole             | \$ 2.18  | \$ 0.07  |                  |
| nystatine                 | \$ 3.87  |          | \$ 3.87          |
| amoxicillin               | \$ 2.42  | \$ 0.22  |                  |
| amoxiclav                 | \$ 20.69 |          | \$ 20.69         |
| ampicillin                | \$ 1.21  |          | \$ 1.21          |
| ceftriaxone               | \$ 2.90  |          | \$ 2.90          |
| chloramphenicol           | \$ 3.15  | \$ 1.45  |                  |
| ciprofloxacin             | \$ 1.94  |          | \$ 1.94          |
| cotrimoxazol              | \$ 1.94  |          | \$ 1.94          |
| doxycycline               |          | \$ 0.73  |                  |
| erythromycin              | \$ 2.42  |          | \$ 2.42          |
| gentamicin                | \$ 1.21  |          | \$ 1.21          |
| penicillin                |          | \$ 0.19  |                  |
| tetracycline              | \$ 0.73  |          | \$ 0.73          |
| cloxacillin               |          | \$ 0.19  |                  |
| <b>Ghana</b>              |          |          |                  |
| aciclovir                 | \$ 3.74  | \$ 0.63  |                  |
| albendazole               | \$ 0.93  | \$ 0.42  |                  |
| artesunate                | \$ 0.99  |          | \$ 0.99          |
| aspirin injectable        | \$ 30.74 |          | \$ 30.74         |

|                           |          |         |          |          |
|---------------------------|----------|---------|----------|----------|
| candiderm cream           | \$ 4.35  |         | \$ 4.35  | [50]     |
| carbocisteine             | \$ 2.56  |         | \$ 2.56  | [49]     |
| cetirizine                | \$ 1.26  | \$ 0.03 |          | [49]     |
| chlorpheniramine          | \$ 1.22  | \$ 0.01 |          | [49]     |
| clotrimazole              | \$ 1.58  |         | \$ 1.58  | [49]     |
| coartem                   | \$ 1.34  |         | \$ 1.34  | [49]     |
| diclofenac                |          | \$ 0.28 |          | [49]     |
| griseofulvin              | \$ 3.02  | \$ 0.20 |          | [49]     |
| ibuprofen                 | \$ 1.18  | \$ 0.02 |          | [49]     |
| linctus                   | \$ 2.25  |         | \$ 2.25  | [49]     |
| mebendazole               | \$ 1.58  |         | \$ 0.71  | [49]     |
| miconazole                | \$ 2.36  |         | \$ 2.36  | [49]     |
| oral rehydration solution |          | \$ 0.30 |          | [49]     |
| paracetamol               | \$ 2.04  | \$ 0.02 |          | [49]     |
| saline                    | \$ 1.18  |         | \$ 1.18  | [49]     |
| syrup apflu               | \$ 7.39  |         | \$ 7.39  | Local PI |
| metronidazole             | \$ 1.18  | \$ 0.03 |          | [49]     |
| nystatine                 | \$ 5.91  | \$ 0.29 |          | [49]     |
| throat lozenges           | \$ 10.43 |         | \$ 10.43 | [51]     |
| amoxicillin               | \$ 1.18  | \$ 0.05 |          | [49]     |
| amoxiclav                 | \$ 5.12  | \$ 0.52 |          | [49]     |
| ampicillin                | \$ 0.39  |         | \$ 0.39  | [49]     |
| azithromycin              | \$ 3.74  | \$ 0.52 |          | [49]     |
| cefixime                  | \$ 5.65  |         | \$ 5.65  | Local PI |
| ceftriaxone               | \$ 2.66  |         | \$ 2.66  | [49]     |
| cefuroxime                |          | \$ 0.43 |          | Local PI |
| chloramphenicol           | \$ 0.99  |         | \$ 0.99  | [49]     |
| ciprofloxacin             | \$ 1.97  | \$ 0.09 |          | [49]     |
| cotrimoxazol              |          | \$ 0.03 |          | [49]     |
| doxycycline               |          | \$ 0.06 |          | [49]     |
| erythromycin              | \$ 1.97  | \$ 0.08 |          | [49]     |
| flucloxacillin            | \$ 1.58  | \$ 0.11 |          | [49]     |
| gentamicin                | \$ 0.79  |         | \$ 0.79  | [49]     |
| mupirocin                 | \$ 3.76  |         | \$ 3.76  | Local PI |
| neohycolex                | \$ 9.13  |         | \$ 9.13  | [52]     |
| nitrofurantoin            |          | \$ 0.71 |          | [49]     |
| penicillin                |          | \$ 0.04 |          | [49]     |
| tetracycline              | \$ 0.75  | \$ 0.03 |          | [49]     |
| <b>Uganda</b>             |          |         |          |          |
| aciclovir                 | \$ 3.04  | \$ 0.02 |          | Local PI |
| albendazole               | \$ 1.52  |         | \$ 1.52  | Local PI |
| artesunate                | \$ 4.56  |         | \$ 4.56  | Local PI |
| aspirin injectable        | \$ 0.05  |         | \$ 0.05  | [53]     |
| candiderm cream           | \$ 6.08  |         | \$ 6.08  | Local PI |
| carbocisteine             | \$ 4.56  |         | \$ 4.56  | [54]     |
| cetirizine                | \$ 6.08  | \$ 0.19 |          | Local PI |
| chlorpheniramine          |          | \$ 0.04 |          | Local PI |

|                           |          |         |          |          |
|---------------------------|----------|---------|----------|----------|
| clotrimazole              | \$ 2.28  | \$ 0.07 |          | Local PI |
| coartem                   | \$ 5.32  |         | \$ 5.32  | Local PI |
| diclofenac                |          | \$ 0.04 |          | Local PI |
| duocotexcin               | \$ 17.48 |         | \$ 17.48 | Local PI |
| fluconazole               |          | \$ 0.46 |          | Local PI |
| griseofulvin              |          | \$ 0.23 |          | [54]     |
| ibuprofen                 | \$ 3.80  | \$ 0.08 |          | Local PI |
| ketoconazole              |          | \$ 0.06 |          | [53]     |
| linctus                   | \$ 4.18  |         | \$ 4.18  | Local PI |
| loperamide                |          | \$ 0.08 |          | Local PI |
| mebendazole               |          | \$ 0.08 |          | Local PI |
| oral rehydration solution |          | \$ 0.38 |          | Local PI |
| paracetamol               | \$ 3.80  | \$ 0.04 |          | Local PI |
| piritex                   | \$ 3.80  | \$ 4.18 |          | Local PI |
| saline                    | \$ 2.28  |         | \$ 2.28  | Local PI |
| syrup apflu               | \$ 5.70  |         | \$ 5.70  | Local PI |
| tranexamic acid           |          | \$ 1.52 |          | Local PI |
| metronidazole             | \$ 4.56  | \$ 0.08 |          | Local PI |
| nystatine                 | \$ 1.20  | \$ 0.23 |          | Local PI |
| throat lozenges           | \$ 0.38  |         | \$ 0.38  | Local PI |
| amoxicillin               | \$ 3.04  | \$ 0.15 |          | Local PI |
| amoxiclav                 | \$ 15.20 | \$ 3.42 |          | Local PI |
| ampicillin                | \$ 3.80  | \$ 0.15 |          | Local PI |
| ampiclox                  | \$ 4.56  | \$ 0.23 |          | Local PI |
| azithromycin              | \$ 6.08  | \$ 0.61 |          | Local PI |
| cefixime                  | \$ 12.92 | \$ 1.52 |          | Local PI |
| ceftriaxone               | \$ 1.90  |         | \$ 1.90  | Local PI |
| cefuroxime                | \$ 6.84  | \$ 1.52 |          | Local PI |
| chloramphenicol           |          | \$ 0.23 |          | Local PI |
| ciprofloxacin             |          | \$ 0.15 |          | Local PI |
| cotrimoxazol              | \$ 3.04  | \$ 0.15 |          | Local PI |
| doxycycline               |          | \$ 0.15 |          | Local PI |
| erythromycin              | \$ 4.56  | \$ 0.23 |          | Local PI |
| flucloxacillin            | \$ 19.76 | \$ 1.90 |          | Local PI |
| gentamicin                | \$ 1.14  |         | \$ 1.14  | Local PI |
| levofloxacin              |          | \$ 0.38 |          | Local PI |
| neohycolex                | \$ 4.56  |         | \$ 4.56  | Local PI |
| nitrofurantoin            |          | \$ 0.08 |          | Local PI |
| penicillin                |          | \$ 0.15 |          | Local PI |
| tetracycline              | \$ 0.49  |         | \$ 0.49  | [53]     |
| zedex                     | \$ 4.56  |         | \$ 4.56  | [54]     |

*Abbreviations: \$, International dollar*

**Table A4** Defined Daily Doses per antibiotic according to the standards as defined by the World Health Organization

| Drug            | Defined Daily Doses (in milligrams) | Source |
|-----------------|-------------------------------------|--------|
| Amoxicillin     | 1500                                | [33]   |
| Amoxiclav       | 1500                                | [33]   |
| Ampicillin      | 2000                                | [33]   |
| Ampiclox        | 2000                                | [33]   |
| Azithromycin    | 300                                 | [33]   |
| Cefixime        | 400                                 | [33]   |
| Ceftriaxone     | 2000                                | [33]   |
| Cefuroxime      | 500                                 | [33]   |
| Chloramphenicol | 3000                                | [33]   |
| Ciprofloxacin   | 1000                                | [33]   |
| Cloxacillin     | 2000                                | [33]   |
| Co-trimoxazole  | 1920                                | [55]   |
| Doxycycline     | 100                                 | [33]   |
| Erythromycin    | 1000                                | [33]   |
| Flucloxacillin  | 2000                                | [33]   |
| Levofloxacin    | 500                                 | [33]   |
| Metronidazole   | 1500                                | [33]   |
| Nitrofurantoin  | 200                                 | [33]   |
| Penicillin*     | 2000                                | [33]   |

\* Defined Daily Doses for phenoxymethylpenicillin is considered.

**Table A5** Test in scope of Standard-of-Care including percentages of patients tested

|                                  | Burkina Faso | Ghana | Uganda |
|----------------------------------|--------------|-------|--------|
| <i>Pathogen-specific</i>         |              |       |        |
| Group A Streptococcus test       | -            | 1.1%  | 0.1%   |
| Influenza test                   | -            | 0.9%  | 0.2%   |
| Malaria RDT/Malaria blood smear  | 100%         | 92.7% | 80.9%  |
| Respiratory syncytial virus test | -            | 0.9%  | -      |
| Streptococcus pneumoniae test    | -            | 0.5%  | 0.2%   |
| Typhoid fever test               | -            | 2.7%  | 0.5%   |
| Widal test                       | -            | 0.4%  | 0.1%   |
| HIV test                         | -            | -     | 0.4%   |
| Blood film test                  | -            | -     | 1.9%   |
| Urine microscopy                 | -            | -     | 3.6%   |
| Stool microscopy                 | -            | -     | 0.2%   |
| <i>Non-pathogen-specific</i>     |              |       |        |
| CRP test                         | 0.1%         | 2.4%  | 0.7%   |
| Urine test                       | -            | 6.1%  | 0.3%   |
| CBC test                         | 0.1%         | 97.7% | 0.7%   |
| Hb test                          | -            | 54.0% | -      |
| No tests performed               | 0%           | 0.5%  | 10.8%  |

Abbreviations: CBC, Complete Blood Count; CRP, C-Reactive Protein; Hb, Hemoglobin; HIV, human immunodeficiency virus.

**Table A6** Overview of costs, segregated per cost category, per strategy and per country

Table A10

Overview of costs, segregated per cost category, per strategy and per country

| Costs (% of total cost) |                 |                |                 |                  |                        |               |               |
|-------------------------|-----------------|----------------|-----------------|------------------|------------------------|---------------|---------------|
| Strategy                | Test            | Therapeutics   | Labour          | Therapeutics OOP | Non-favourable outcome | Training      | Equipment     |
| Uganda                  |                 |                |                 |                  |                        |               |               |
| Trial-based             | \$ 39.95 (70.5) | \$ 5.78 (10.2) | \$ 9.07 (16.0)  | \$ 0.05 (0.1)    | \$ 1.31 (2.3)          | \$ 0.13 (0.2) | \$ 0.37 (0.7) |
| Malaria                 | \$ 5.75 (36.3)  | \$ 5.04 (31.8) | \$ 3.36 (21.2)  | \$ 0.06 (0.4)    | \$ 1.50 (9.5)          | \$ 0.13 (0.8) | \$ 0.00       |
| Malaria + CRP           | \$ 7.37 (37.6)  | \$ 4.58 (23.4) | \$ 5.46 (27.8)  | \$ 0.06 (0.3)    | \$ 1.90 (9.7)          | \$ 0.13 (0.7) | \$ 0.11 (0.6) |
| Malaria + CBC           | \$ 14.37 (53.6) | \$ 4.60 (17.2) | \$ 5.46 (20.4)  | \$ 0.06 (0.2)    | \$ 1.94 (7.2)          | \$ 0.13 (0.5) | \$ 0.25 (0.9) |
| Malaria + CRP + CBC     | \$ 15.98 (56.2) | \$ 4.52 (15.9) | \$ 5.46 (19.2)  | \$ 0.06 (0.2)    | \$ 1.93 (6.8)          | \$ 0.13 (0.5) | \$ 0.37 (1.3) |
| Malaria + All           | \$ 25.27 (60.5) | \$ 4.98 (11.9) | \$ 9.07 (21.7)  | \$ 0.06 (0.1)    | \$ 1.89 (4.5)          | \$ 0.13 (0.3) | \$ 0.37 (0.9) |
| Standard-of-Care        | \$ 5.69 (35.6)  | \$ 5.88 (36.8) | \$ 3.36 (21.0)  | \$ 0.04 (0.3)    | \$ 1.02 (6.4)          | \$ 0.00       | \$ 0.00       |
| Ghana                   |                 |                |                 |                  |                        |               |               |
| Trial-based             | \$ 53.15 (76.0) | \$ 4.84 (6.9)  | \$ 11.48 (16.4) | \$ 0.01 (0.0)    | \$ 0.16 (0.2)          | \$ 0.15 (0.2) | \$ 0.11 (0.2) |
| Malaria                 | \$ 6.17 (38.8)  | \$ 4.40 (27.6) | \$ 4.62 (29.0)  | \$ 0.01 (0.1)    | \$ 0.57 (3.6)          | \$ 0.15 (0.9) | \$ 0.00       |
| Malaria + CRP           | \$ 8.69 (40.9)  | \$ 3.80 (17.9) | \$ 7.51 (35.3)  | \$ 0.01 (0.0)    | \$ 0.98 (4.6)          | \$ 0.15 (0.7) | \$ 0.11 (0.5) |
| Malaria + CBC           | \$ 15.59 (54.1) | \$ 4.86 (16.9) | \$ 7.51 (26.1)  | \$ 0.01 (0.0)    | \$ 0.68 (2.4)          | \$ 0.15 (0.5) | \$ 0.00       |
| Malaria + CRP + CBC     | \$ 18.11 (59.0) | \$ 3.78 (12.3) | \$ 7.51 (24.5)  | \$ 0.01 (0.0)    | \$ 0.99 (3.2)          | \$ 0.15 (0.5) | \$ 0.11 (0.4) |
| Malaria + All           | \$ 46.82 (74.1) | \$ 4.18 (6.6)  | \$ 11.48 (18.2) | \$ 0.01 (0.0)    | \$ 0.43 (0.7)          | \$ 0.15 (0.2) | \$ 0.11 (0.2) |
| Standard-of-Care        | \$ 17.72 (64.9) | \$ 4.79 (17.5) | \$ 4.62 (16.9)  | \$ 0.01 (0.0)    | \$ 0.18 (0.7)          | \$ 0.00       | \$ 0.00       |
| Burkina Faso            |                 |                |                 |                  |                        |               |               |
| Trial-based             | \$ 25.89 (59.9) | \$ 7.62 (17.6) | \$ 8.97 (20.7)  | \$ 0.00          | \$ 0.13 (0.3)          | \$ 0.27 (0.6) | \$ 0.37 (0.9) |
| Malaria                 | \$ 1.14 (9.8)   | \$ 7.19 (62.0) | \$ 2.75 (23.7)  | \$ 0.00          | \$ 0.26 (2.2)          | \$ 0.27 (2.3) | \$ 0.00       |
| Malaria + CRP           | \$ 1.81 (12.9)  | \$ 7.09 (50.4) | \$ 4.46 (31.7)  | \$ 0.00          | \$ 0.32 (2.3)          | \$ 0.27 (1.9) | \$ 0.11 (0.8) |
| Malaria + CBC           | \$ 4.56 (26.8)  | \$ 7.18 (42.2) | \$ 4.46 (26.2)  | \$ 0.00          | \$ 0.30 (1.8)          | \$ 0.27 (1.6) | \$ 0.25 (1.5) |
| Malaria + CRP + CBC     | \$ 5.24 (29.5)  | \$ 7.09 (40.0) | \$ 4.46 (25.1)  | \$ 0.00          | \$ 0.31 (1.7)          | \$ 0.27 (1.5) | \$ 0.37 (2.1) |
| Malaria + All           | \$ 8.10 (32.2)  | \$ 7.09 (28.2) | \$ 8.97 (35.7)  | \$ 0.00          | \$ 0.36 (1.4)          | \$ 0.27 (1.1) | \$ 0.37 (1.5) |
| Standard-of-Care        | \$ 1.16 (9.5)   | \$ 8.22 (67.1) | \$ 2.75 (22.5)  | \$ 0.00          | \$ 0.12 (1.0)          | \$ 0.00       | \$ 0.00       |

Abbreviations: CBC, complete blood count; CRP, C-reactive protein; \$, International dollar; OOP, out-of-pocket.

**Table A7** Overview of secondary outcome measures

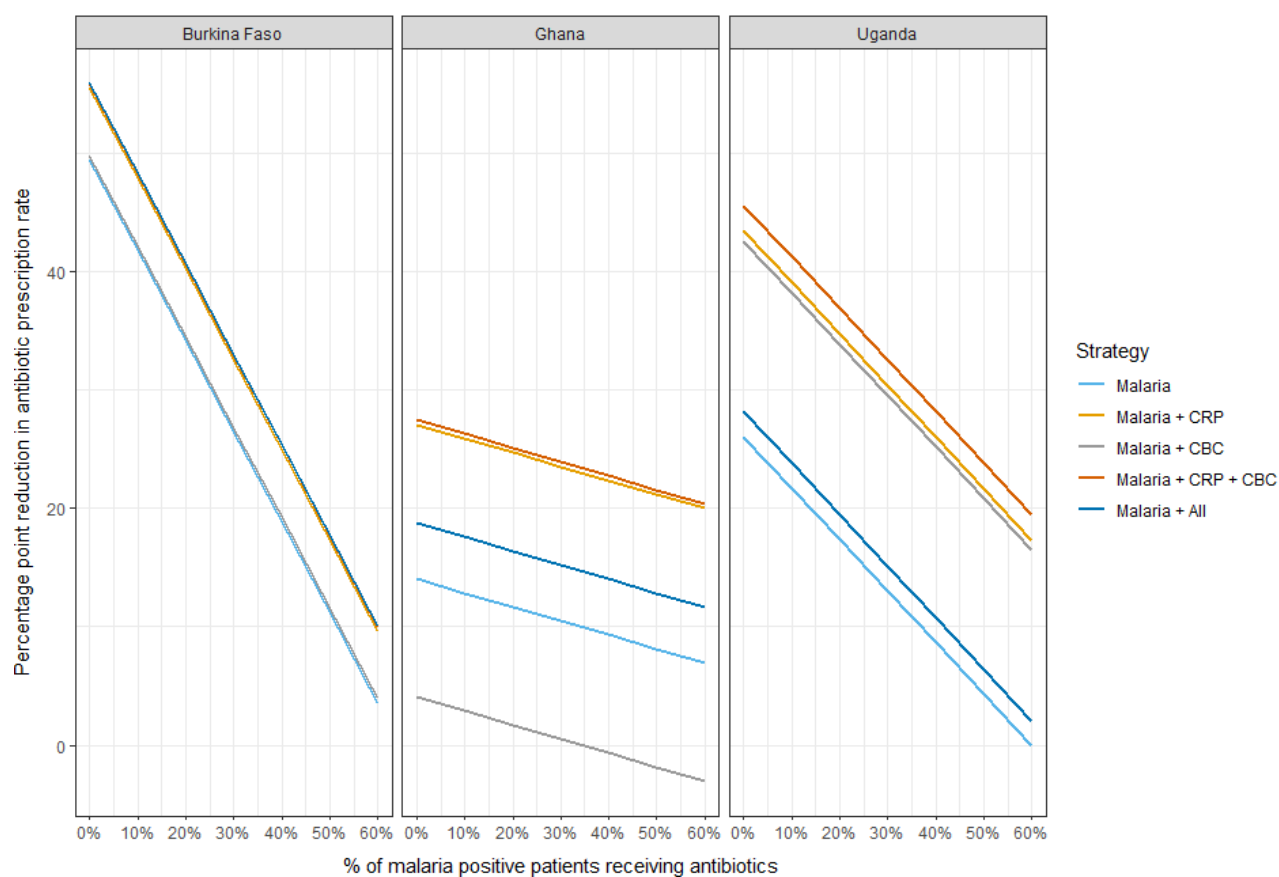
| Scenario            | Number of antibiotic regimen consumed | Defined Daily Doses | Days-of-Therapy | DALY      |
|---------------------|---------------------------------------|---------------------|-----------------|-----------|
| <b>Uganda</b>       |                                       |                     |                 |           |
| Trial-based         | 0.62                                  | 3.05                | 3.57            | 0.0000622 |
| Malaria             | 0.32                                  | 1.77                | 1.89            | 0.0001229 |
| Malaria + CRP       | 0.11                                  | 0.51                | 0.60            | 0.0001460 |
| Malaria + CBC       | 0.12                                  | 0.57                | 0.66            | 0.0001487 |
| Malaria + CRP + CBC | 0.08                                  | 0.38                | 0.45            | 0.0001484 |
| Malaria + All       | 0.30                                  | 1.64                | 1.74            | 0.0000975 |
| Standard-of-Care    | 0.66                                  | 2.85                | 3.34            | 0.0000846 |
| <b>Ghana</b>        |                                       |                     |                 |           |
| Trial-based         | 0.38                                  | 1.03                | 2.36            | 0.0000095 |
| Malaria             | 0.28                                  | 0.74                | 1.73            | 0.0000697 |
| Malaria + CRP       | 0.14                                  | 0.37                | 0.88            | 0.0000884 |
| Malaria + CBC       | 0.39                                  | 1.02                | 2.38            | 0.0000610 |
| Malaria + CRP + CBC | 0.14                                  | 0.36                | 0.86            | 0.0000896 |
| Malaria + All       | 0.23                                  | 0.61                | 1.43            | 0.0000283 |
| Standard-of-Care    | 0.42                                  | 1.22                | 2.57            | 0.0000102 |
| <b>Burkina Faso</b> |                                       |                     |                 |           |
| Trial-based         | 0.39                                  | 1.06                | 2.58            | 0.0000028 |
| Malaria             | 0.12                                  | 0.31                | 0.81            | 0.0000159 |
| Malaria + CRP       | 0.06                                  | 0.15                | 0.40            | 0.0000194 |
| Malaria + CBC       | 0.12                                  | 0.30                | 0.78            | 0.0000179 |
| Malaria + CRP + CBC | 0.06                                  | 0.14                | 0.37            | 0.0000190 |
| Malaria + All       | 0.05                                  | 0.14                | 0.37            | 0.0000169 |
| Standard-of-Care    | 0.66                                  | 1.59                | 4.39            | 0.0000027 |

Abbreviations: CBC, complete blood count; CRP, C-reactive protein; DALY, disability-adjusted life-years.

**Table A8** Results of scenario analysis

|                     | Non-respiratory patient population |                              |                                    |                  | Respiratory patient population |                              |                                    |                  |
|---------------------|------------------------------------|------------------------------|------------------------------------|------------------|--------------------------------|------------------------------|------------------------------------|------------------|
|                     | Cost                               | Antibiotic prescription rate | Percentage point reduction in APR* | Cost difference* | Cost                           | Antibiotic prescription rate | Percentage point reduction in APR* | Cost difference* |
| <b>Uganda</b>       |                                    |                              |                                    |                  |                                |                              |                                    |                  |
| Malaria             | \$ 15.41                           | 25.0%                        | 26.0%                              | Dominant         | \$ 16.38                       | 24.1%                        | 26.1%                              | Dominant         |
| Standard-of-Care    | \$ 15.62                           | 51.1%                        | -                                  | -                | \$ 16.45                       | 50.2%                        | -                                  | -                |
| Malaria + CRP       | \$ 18.77                           | 3.3%                         | 47.8%                              | \$ 3.15          | \$ 20.63                       | 12.1%                        | 38.1%                              | \$ 4.18          |
| Malaria + CBC       | \$ 25.14                           | 4.0%                         | 47.0%                              | \$ 9.52          | \$ 28.87                       | 13.1%                        | 37.2%                              | \$ 12.42         |
| Malaria + CRP + CBC | \$ 26.58                           | 1.5%                         | 49.5%                              | \$ 10.96         | \$ 30.73                       | 9.5%                         | 40.7%                              | \$ 14.28         |
| Malaria + All       | \$ 36.10                           | 23.1%                        | 27.9%                              | \$ 20.48         | \$ 48.67                       | 21.8%                        | 28.4%                              | \$ 32.22         |
| Trial-based         | \$ 50.91                           | 57.2%                        | -6.2%                              | \$ 35.29         | \$ 63.63                       | 50.3%                        | -0.1%                              | \$ 47.18         |
| <b>Ghana</b>        |                                    |                              |                                    |                  |                                |                              |                                    |                  |
| Malaria             | \$ 16.25                           | 12.0%                        | 20.8%                              | Dominant         | \$ 15.66                       | 37.3%                        | 8.9%                               | Dominant         |
| Malaria + CRP       | \$ 21.56                           | 3.7%                         | 29.1%                              | Dominant         | \$ 21.02                       | 20.7%                        | 25.5%                              | Dominant         |
| Standard-of-Care    | \$ 26.72                           | 32.7%                        | -                                  | -                | \$ 27.78                       | 46.3%                        | -                                  | -                |
| Malaria + CBC       | \$ 28.02                           | 24.3%                        | 8.5%                               | \$ 1.30          | \$ 29.38                       | 45.5%                        | 0.7%                               | \$ 1.60          |
| Malaria + CRP + CBC | \$ 29.84                           | 3.3%                         | 29.4%                              | \$ 3.12          | \$ 31.30                       | 20.2%                        | 26.0%                              | \$ 3.52          |
| Malaria + All       | \$ 59.32                           | 15.0%                        | 17.8%                              | \$ 32.60         | \$ 66.13                       | 26.7%                        | 19.5%                              | \$ 38.36         |
| Trial-based         | \$ 71.96                           | 33.6%                        | -0.8%                              | \$ 45.24         | \$ 68.33                       | 37.8%                        | 8.4%                               | \$ 40.55         |
| <b>Burkina Faso</b> |                                    |                              |                                    |                  |                                |                              |                                    |                  |
| Malaria             | \$ 11.45                           | 6.5%                         | 20.7%                              | Dominant         | \$ 11.73                       | 16.5%                        | 74.7%                              | Dominant         |
| Standard-of-Care    | \$ 11.96                           | 27.2%                        | -                                  | -                | \$ 12.50                       | 91.2%                        | -                                  | -                |
| Malaria + CRP       | \$ 13.72                           | 0.9%                         | 26.3%                              | \$ 1.76          | \$ 14.37                       | 10.0%                        | 81.2%                              | \$ 1.87          |
| Malaria + CBC       | \$ 15.79                           | 5.9%                         | 21.3%                              | \$ 3.83          | \$ 18.13                       | 16.3%                        | 74.9%                              | \$ 5.63          |
| Malaria + CRP + CBC | \$ 16.31                           | 0.6%                         | 26.6%                              | \$ 4.35          | \$ 18.99                       | 9.6%                         | 81.6%                              | \$ 6.49          |
| Malaria + All       | \$ 22.06                           | 0.6%                         | 26.6%                              | \$ 10.10         | \$ 27.90                       | 9.4%                         | 81.8%                              | \$ 15.40         |
| Trial-based         | \$ 39.79                           | 23.3%                        | 3.9%                               | \$ 27.83         | \$ 46.29                       | 50.9%                        | 40.3%                              | \$ 33.79         |

Abbreviations: APR, antibiotic prescription rate; CBC, complete blood count; CRP, C-reactive protein; SoC, Standard-of-Care; \$, International dollar. \*compared to SOC



**Fig. A1** Scenario analysis: impact of alternative percentages of malaria positive patients prescribed antibiotics on percentage point reduction of APR in a pairwise comparison with Standard-of-Care

Abbreviations: CBC, complete blood count; CRP, C-reactive protein.